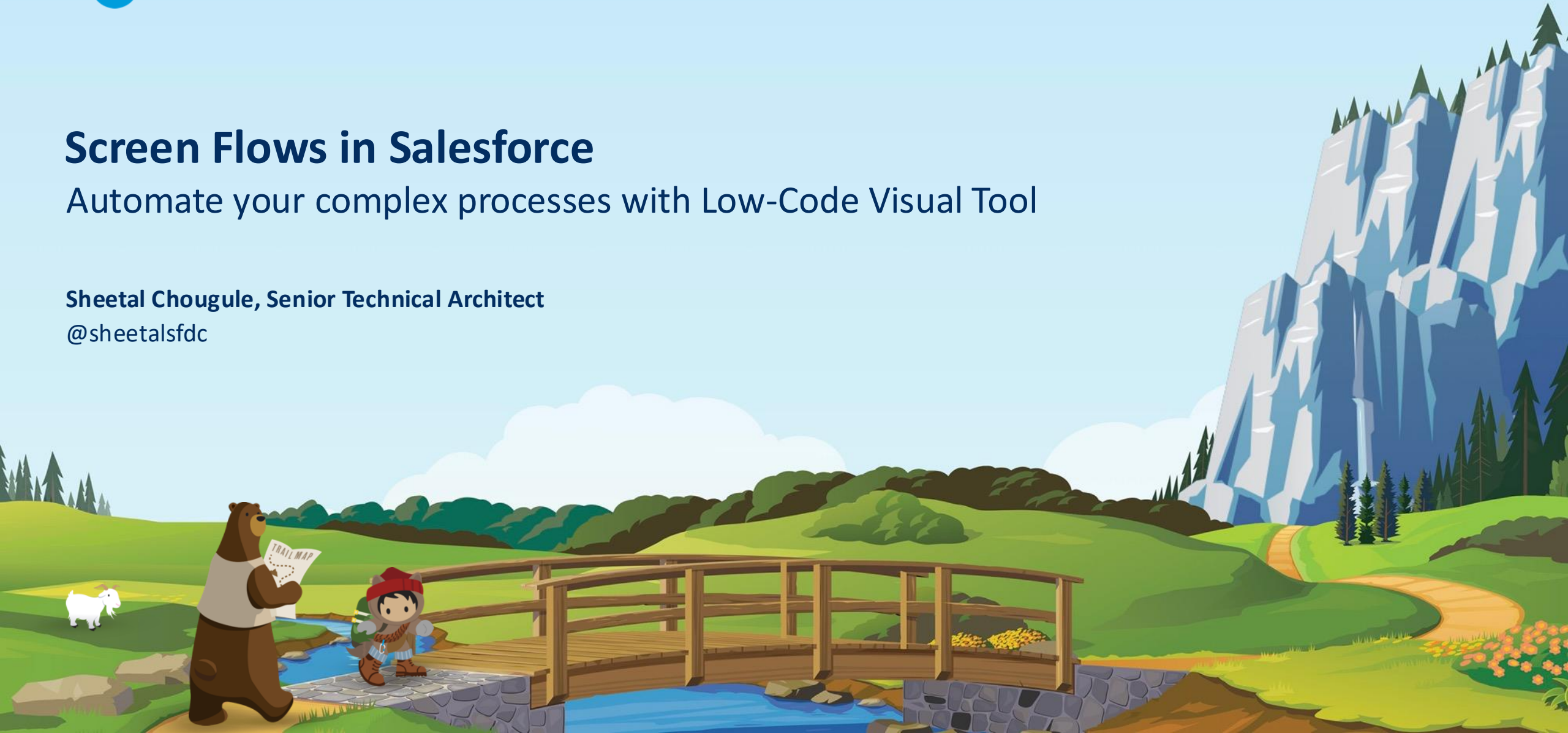


# Screen Flows in Salesforce

Automate your complex processes with Low-Code Visual Tool

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@sheetalsfdc



# Ground Rules

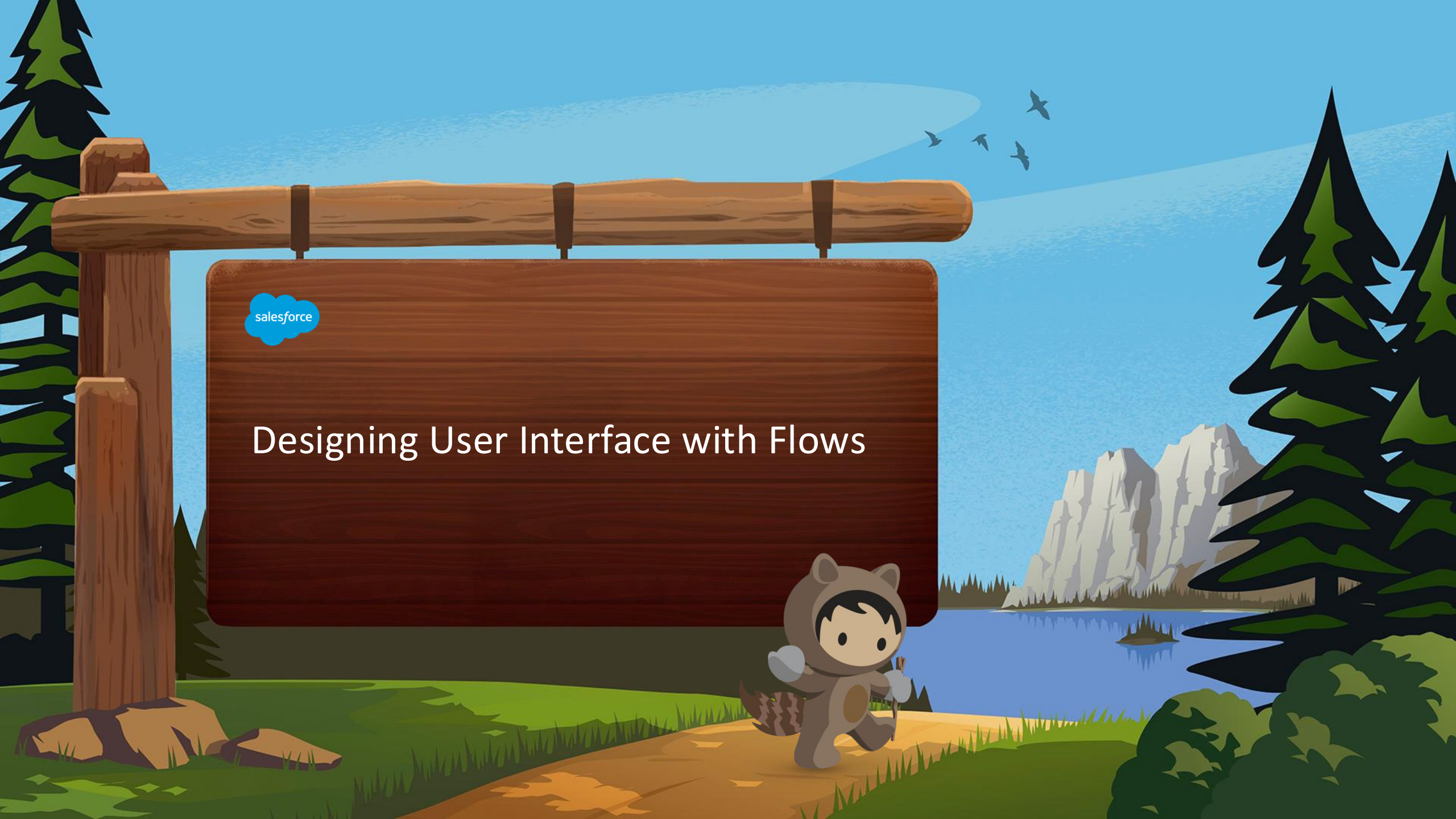
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Let's agree to some ground rules before we proceed

- Training will be delivered considering you do not know anything about Salesforce. So, even though you have prior experience keep your mind blank and open to learn more effectively.
- This is online live Training Session. Focus completely on session, even though recordings will be made available.
- Please make sure there is No background noise or disturbances.
- Questions will be addressed at the end of the session and will also setup a recurring meeting to discuss that week's question.
- **NO QUESTION IS SILLY.** So, you can ask same question again and again till you understand completely.
- **Phones on Silent.**
- **Complete all the assignments on time so you can understand next session.**
- **Do not forward invites unless you confirm with Trainer first.**



Keep it  
Interactive



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# Designing User Interface with Flows

# Screen Flows



Adding visual interaction for users, for gathering inputs, displaying list of records for selection or creation of new records or perform similar actions

- Invoke a screen flow from many distribution methods, such as
  - Lightning pages,
  - Experience Builder pages,
  - Lightning web components,
  - Aura components,
  - custom buttons, actions or links,
  - Flow Orchestration
- Choose from a set of input fields (Date/Time, Text, Long Text Area, Lookup, just to name a few) and a lot of components (such as File Upload, Slider, Toggle, Slack Workplace Selector, Display Image, Slider, or Display Text).
- Component visibility is a feature that allows you to specify when to show a component if conditions you configure are true.
- The Screen element can display data to users or collect information from them.
- Flows can interact with users by presenting questions and information on a screen, or by sending various types of communications.
- The Screen element and the Action element make these interactions possible.

A screenshot of a Salesforce 'New Contact' screen flow. The form is titled 'New Contact' and contains several input fields: 'First Name' and 'Last Name' are text inputs; 'Account' is a dropdown menu with 'Acme' selected. Below the dropdown is a toggle switch labeled 'If this contact already exists, update the existing record.' with the text 'Create other contact' underneath. A blue 'Next' button is located at the bottom right of the form.

**New Contact**

Name

First Name

Last Name

\* Account

Acme

If this contact already exists, update the existing record. ☐ Create other contact

Next



let's get to  
**WORK**



## Build a Calculator using Screen Flow

- Take input from user in the fields – Num1 & Num2
- Create dropdown to select the Operations – Add, Subtract, Multiply, Divide
- Based on the Operation selected – Calculate result and display on the next screen.



# Flow Resources



# Everything is Resource in flow

Build resources to store, process and display data throughout the flow.

Resource Type

Select...

Variable

Constant

Formula

Text Template

Choice

Record Choice Set

Picklist Choice Set

Stage

Text

Record

Number

Currency

Boolean

Date

Date/Time

Picklist

Multi-Select Picklist

Select...

Availability Outside the Flow

☐ Available for input

☐ Available for output

Select the type of data you want to store in the resource.

Pass the values into variable. Enables to receive input while invocation or exposes the flow variables to other processes.

☐ Allow multiple values (collection) 

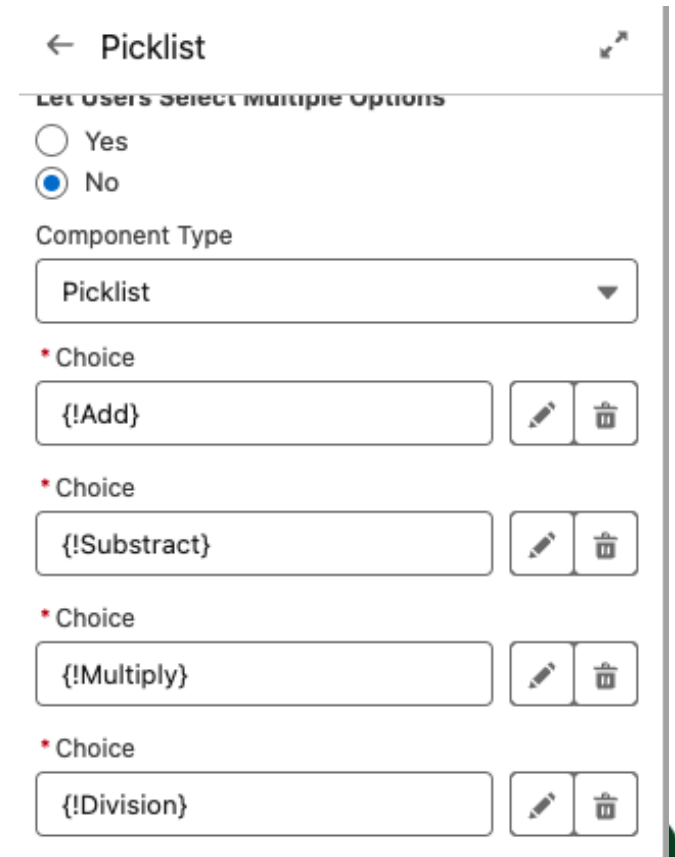
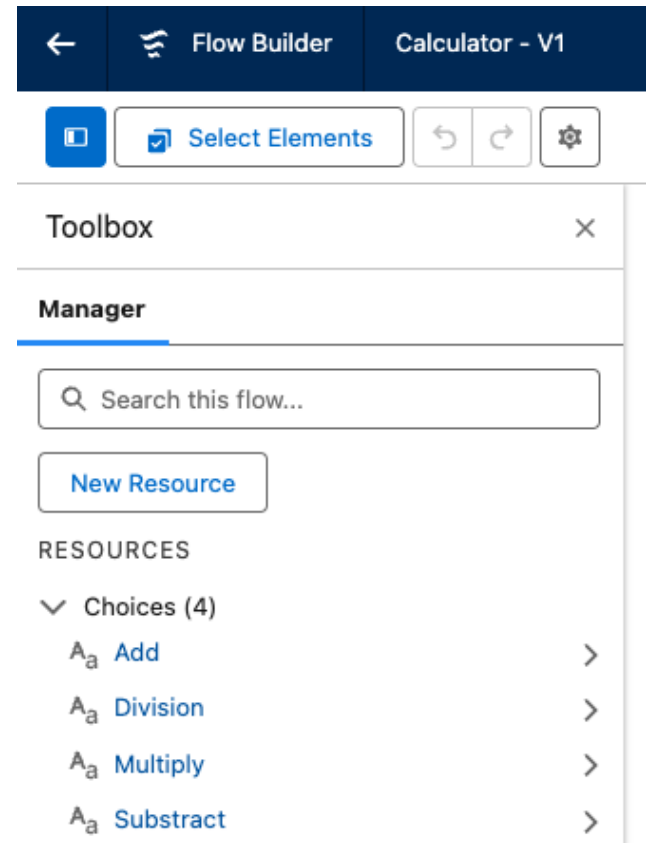
Store multiple values in the resource

# Choice

Choice – A single value defined in the flow the user can choose.



- Choices are used as options that users can select from within radio buttons or multi-select picklist components.
- composed of both a label and a value, which are often confused – the **label** is displayed to end users, while the **value** is what is stored behind the scenes and used by the Flow.
- Allows to create 1 picklist choice at a time.
- Values are stored as individual choice items. But can be used in any Picklist field.
- Make sure to use the correct naming for the



# Types of Choices

Flows enable to create choices based on static or dynamic values

## 1. Collection Choice Set

- group of options made up of existing Collection Variables used in your Flow.
- can only reference Collections that already exist, unlike Record Choice Sets and Picklist Choice Sets that are constructed when they are configured.

## 2. Record Choice Set

- While the Record Choice Set behaves similarly to the Collection Choice Set, it is populated using a query built into the set itself, whereas a Collection Choice Set uses a collection that already exists within the Flow.

## 3. Picklist Choice Set

- Picklist Choice Sets use Picklist fields that are configured on objects to provide options within radio buttons or multi-select picklist Screen components.



# Creating New Choice



**New Resource**

\* Resource Type  
Choice

\* API Name  
Other\_Op

Description

\* Choice Label  
Resource Picker  
Insert a resource...  
Other Operation

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\* Data Type  
Text

Choice Value  
Other Operation

☒ Display text input

**Configure Text Input**

\* Input Label  
Select Other Operation to be added

☐ Require  
☐ Validate

Cancel Done

**Calculator**

\* Input Value 1  
0

\* Input Value 2  
0

\* Operation  
--None--  
Add  
Subtract  
Multiply  
Division  
Other Operation

**Calculator**

\* Input Value 1  
0

\* Input Value 2  
0

\* Operation  
Other Operation  
Select Other Operation to be added

Result  
0

- If we select the “Display text input”, on selection of the picklist choice the new input text is added.
- This input field can be marked as required.
- Validate can also enforce data validation on the field.



# Collection Choice Set



Use an existing collection of records or external data to generate a set of choices. Collection choice sets are useful when a flow reuses the same dataset over multiple screens.

New Resource

\* Resource Type  
Collection Choice Set

\* API Name  
Accounts\_list

Description

\* Collection  
Search record and Apex-defined collections...

RECORD COLLECTION VARIABLES

Accounts from Get\_Accounts Account

Create list of choices from Records from flow and then the Choice set can be used on Screen Flow and other components

\* Collection  
{!Get\_Accounts}

Configure Each Choice

The flow creates a choice for each item in the collection. Identify which fields to use for each choice's label and value.

\* Choice Label  
Name

\* Data Type  
Text

\* Choice Value  
Id

Cancel Done

## Display Accounts List

### Accounts to Select

- ☐ Test Account-Platinum
- ☐ Genentech Consulting
- ☐ test
- ☐ New Account Source
- ☐ Sample Account for Entitlements
- ☐ GenePoint-Platinum
- ☐ United Oil & Gas, UK
- ☐ United Oil & Gas, Singapore
- ☐ Edge Multiskills-Platinum
- ☐ Burlington Textiles Corp of America-Platinum
- ☐ Pyramid Construction Inc.
- ☐ Dickenson plc
- ☐ Grand Hotels & Resorts Ltd-Gold
- ☐ Express Logistics and Transport
- ☐ University of Arizona
- ☐ United Oil & Gas Corp.
- ☐ Test Account-Platinum

Screen to display records from Choice Collection

# Lookup Component

search for and select one or more records in a flow.

- The Lookup flow screen component isn't compatible with mobile devices or standalone Aura apps.
- Dependent lookup filters aren't supported.
- If the Maximum Selections value is 1 and the Record ID Collection and Record ID are both changed, the Record ID takes precedence. The Record ID Collection is ignored.
- Field API Name and Object API Name are case-sensitive.
- The Lookup flow screen component doesn't support filtering by the source object record type.



# Approach 1: Use Lookup field from Object



Drag and drop the lookup component on the screen.

- **API Name:** Component API Name
- **Field API Name:**
  - The API name of a lookup field on the source object referenced in Object API Name.
  - The lookup field referenced in Field API Name must be a field on the object referenced in Object API Name.
- **Label:**
  - The text displayed at the top of the component
- **Object API Name:**
  - The API name of the source object that has the lookup field referenced in Field API Name.
  - The source object can be any object that has the type of lookup field that you want to use.
  - The lookup field referenced in Field API Name must be a field on the object referenced in Object API Name.
- **Record Id:** Default record to be displayed in selection

no preview is available for this component.

Lookup

Account

Search Accounts...

← Lookup

\* API Name ⓘ

AccountSearch

\* Field API Name ⓘ

AccountId

\* Label ⓘ

Account

\* Object API Name ⓘ

Opportunity

Disabled ⓘ

{!\$GlobalConstant.False}

**API of Field from Object**

# Approach 2: Use Lookup field from Record Variable

Object field directly dragged based on the “Record Variable”. Properties are not editable. User can set the Field Visibility.

Components

Fields

ⓘ Add record fields to your screen. [More Info](#)

Record Variable

varOpportunity

×

Search Opportunity fields...

🔍

✓ Required Fields (3)

📅 Close Date

Ⓐ Name

☰ Stage

✓ Fields (21)

🔍 Account ID

🔍 Campaign ID

Ⓐ Closure Reason

🔍 Contact ID

Search Accounts...

Account ID

Field

+

🗑

🔍 Account ID

No preview is available for this component.

← Field Information

Object:

Opportunity

Field Label:

Account ID

Field Name:

AccountId

Data Type:

Lookup(Account)

Required:

No

Update Compatible:

Yes

Create Compatible:

Yes

ⓘ This field is managed in Object Manager. Your permissions affect the Update Compatible and Create Compatible values displayed here, but the running user's permissions control their access at run time.

[Take me to Object Manager](#)



| Flow Resource         | Summary                                                                                                                                                                                                                                                                                                                    |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Constant</b>       | Variable with static data value. Value of constants cannot be changed in flow.                                                                                                                                                                                                                                             |
| <b>Formulas</b>       | <p>Like field formulas use the functions and use them as merge fields. recalculated every time they're accessed.</p> <p>A flow formula can contain up to 3,900 characters.</p>                                                                                                                                             |
| <b>Text Templates</b> | <p>store a larger block of text, or maybe you need that text to be formatted in a specific way.</p> <p>Use a text template to store the body of an email or a chunk of formatted text to reuse on multiple screens.</p> <p>Like formulas, text templates can also use variables and screen components as merge fields.</p> |





# Best Practices



- ✓ Plan out a flow before implementing it. Maybe use draw.io or lucidchart or any other similar tool to line out the criteria's and actions.
- ✓ DO NOT HARD CODE ANY ID's. Use Get or merge fields from URL to pass the data. Global variables should be used whenever possible.
- ✓ Flows also follow Governor Limits. So, avoid DML's inside a loop element. Perform bulk DML operations on records.
- ✓ By default, the flow shows an error message to the user and emails the admin who created the flow.
- ✓ While creating new version, add description with changes you made so it's to go back to version if needed.
- ✓ Make sure that you test as many possibilities as you can think of before you distribute the flow to your users.
- ✓ Follow naming conventions mentioned on the resources section of this deck.
- ✓ Build your incrementally so you can debug easily in case issue occurs.



The Salesforce logo, consisting of the word "salesforce" in white lowercase letters inside a blue cloud-like shape.

# Screen Flows with Variables



# Quick Create Account and Contact record

Create new screen flow and launch it from Home Page

1. Build screen flow to input – Account and Contact Fields. Mark fields required as needed.
2. Display Contact's Mobile Phone in read only mode. If Mobile is blank display Phone field value.
3. Instead of Name fields for Firstname and Lastname, modify the flow to use the "Name" field.
4. Before creating a new Contact in Salesforce, check for duplicate based on first name, last name and email.
5. If record exist,
  - Display the existing contact record on screen input
  - Display toggle for confirming if existing record should be updated or new record should be created.
  - If the toggle result says "Update existing Record" - update the existing record for contact. Else create new Contact record and link to newly created Account
6. If record does not exist, create new contact and link to the newly created Account.
7. Create a new text template for flow errors and use it.

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**Contact Name:** {!Contact.Name}  
**Contact Id:** {!Contact.Id}  
**Account Id:** {!Contact.AccountId}  
**Account Name:** {!Contact.Account.Name}

**ERROR**  
{!\$Flow.FaultMessage}

# Design the Flow

Before you start building the flow, plan the flow and elements you may need to use.

| Requirement                                                                                                                                                                                                                                                  | Element Type to Use                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Collect information from user (contact's first name, last name, and account) and ask what to do if a matching contact exists.                                                                                                                                | Interaction (Screen)                               |
| Look for a matching contact record based on FirstName, LastName and Email.                                                                                                                                                                                   | Data (Get Records)                                 |
| Check if a matching record was found and follow the corresponding path:                                                                                                                                                                                      | Logic (Decision)                                   |
| <ul style="list-style-type: none"><li>If no match exists, create the contact.</li></ul>                                                                                                                                                                      | Data (Create Records)                              |
| <ul style="list-style-type: none"><li>If a match exists, display the contact details from record and add a toggle:<ul style="list-style-type: none"><li>toggle = true, update that contact.</li><li>toggle = false, create that contact.</li></ul></li></ul> | Data (Update Records)<br><br>Data (Create Records) |
| Rejoin the branches together and then confirm what the flow did in Chatter.                                                                                                                                                                                  | Interaction (Action)                               |
| Confirm that the flow is done.                                                                                                                                                                                                                               | Interaction (Screen)                               |



# Opportunity Creation – Fast Create

Explore how Lookup element works

1. Enable Users to enter the Opportunity Name and Close Date.
2. Select the Accounts to be associated using lookup.
3. Create new Opportunity.

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**Contact Name:** {!Contact.Name}  
**Contact Id:** {!Contact.Id}  
**Account Id:** {!Contact.AccountId}  
**Account Name:** {!Contact.Account.Name}

**ERROR**

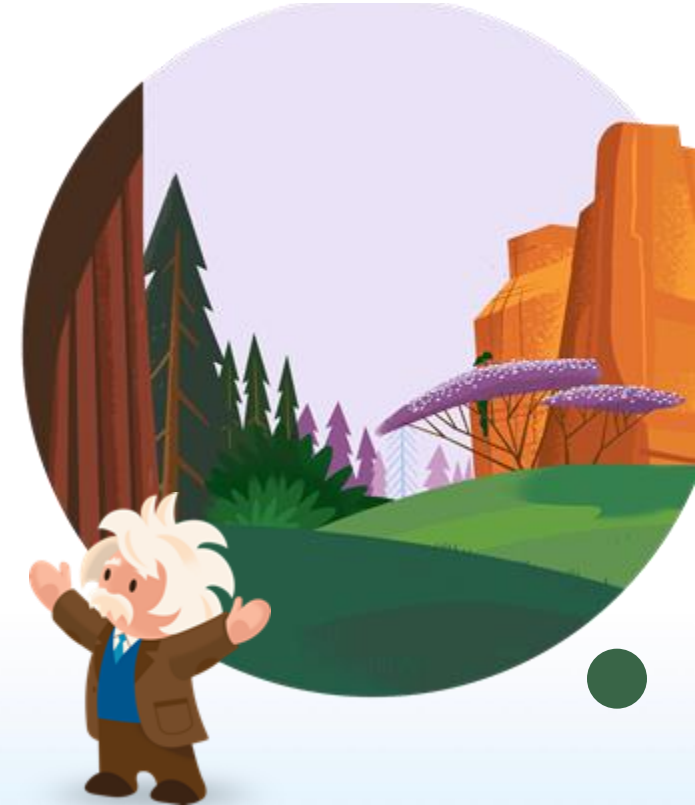
{!\$Flow.FaultMessage}

# Homework

## Building Screen Flows

- Complete homework from the “Homework” folder
  - Screen Flow - Homework
  - Customer Feedback and Issue Resolution
- Complete the Use Case – Display list of Accounts on Screen

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# Resources

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## Naming Conventions:

- <https://wiki.sfxd.org/books/best-practices/page/flow-naming-conventions>
- <https://automationchampion.com/2021/11/09/naming-conventions-for-salesforce-flow/>
- <https://admin.salesforce.com/blog/2021/the-ultimate-guide-to-flow-best-practices-and-standards>

## Trailheads to Complete:

- [Build Flows with Flow Builder](#)
- [\\*\\* Screen Flow Fundamentals Superbadge](#)

Thank You

